



جامعة الطائف
TAIF UNIVERSITY

Technical Report

Robotic Project

Design & Implementation of a 4-Dof Pick-&-Place Robotic Arm

Submitted by:
Meshal Talal AlMehmady
ID: 44102105

• Abstract

This Project Presents the design and Implementation of a 4-Dof robotic arm capable of performing automated Pick-and-Place operations. The system is controlled using an Arduino Uno and PCA9685 PWM Servo driver. A dedicated 5V 3A external power supply is used to ensure stable operation and prevent voltage drops during servo movement. Experimental testing demonstrated reliable and repeatable performance during multiple operating cycles.

• Introduction

Robotic manipulators play an important role in industrial and laboratory automation. They improve efficiency, reduce human effort, and provide consistent operation in repetitive tasks. This project focuses on developing a compact robotic arm capable of performing automated pick-and-place operations using servo motor control and Arduino-based programming.

• Objectives

- Perform automated pick-and-place operations.
- Achieve reliable and repeatable robotic arm motion.
- Ensure stable servo operation using an external power supply.
- Demonstrate a low-cost robotic automation solution.

• System components & cost

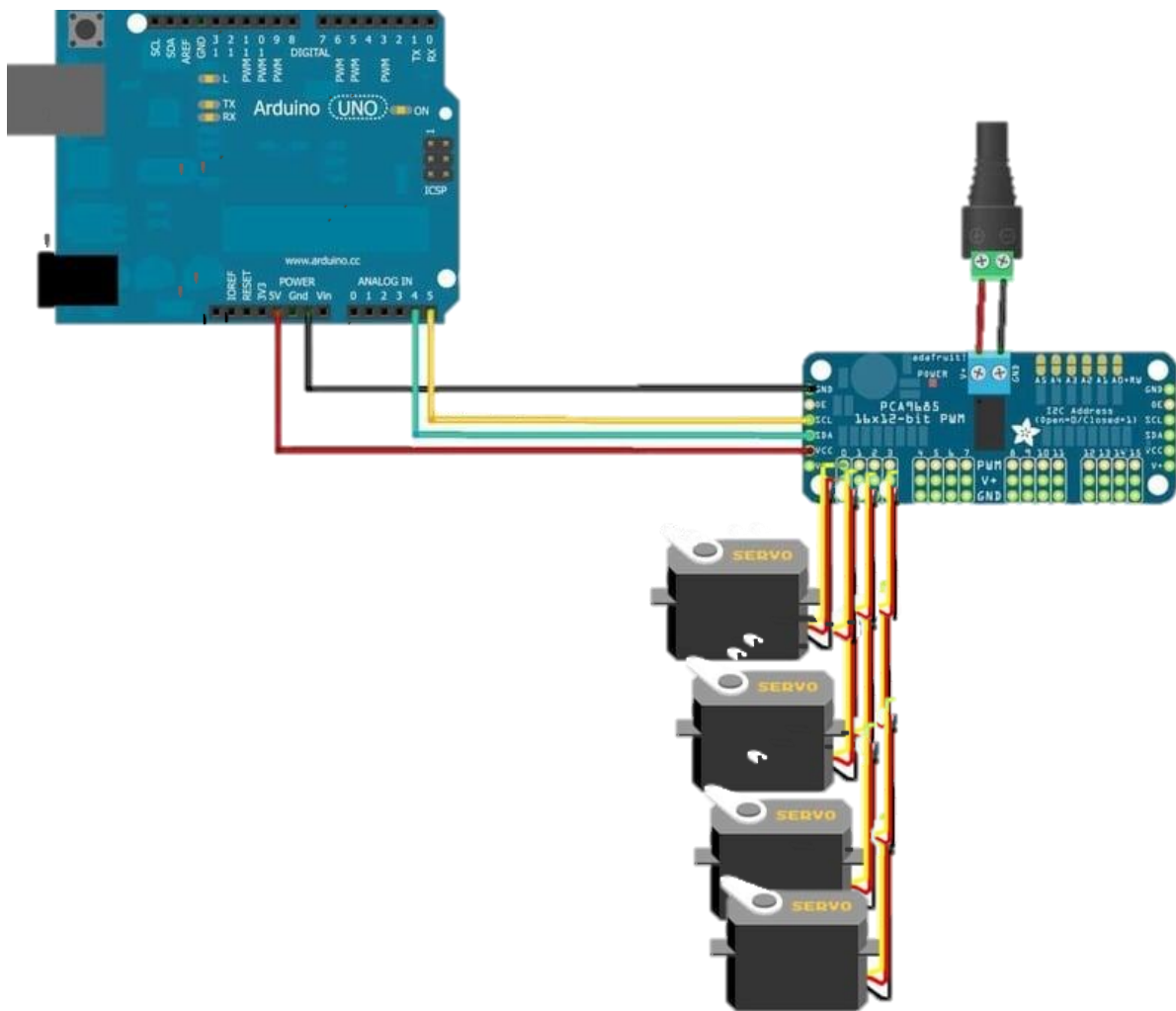
components	cost
Arduino Uno R3	55 SAR
PCA9685 Servo Driver	56 SAR
Four servo Motors	65 SAR
5V 3A External power Adapter	17.25 SAR
4-DoF Robotic Arm Frame	40 SAR
Jumper wires	9 SAR

- System Diagram and Motion Sequence

The Arduino uno communicates with the PCA9685 servo driver through the I2C Protocol using SDA and SCL connections. A common ground is shared between the Arduino and the external power supply to ensure proper signal transmission.

Motion Sequence

- 1- Home position.
- 2- object Pick position.
- 3- object Placement position.
- 4- Return to Home position.



• Software Implementation

PROJECT_44102105.ino

```
1  /*Meshal Talal Al Mehmady
2  ID: 44102105 */
3  #include <Wire.h>
4  #include <Adafruit_PWMServoDriver.h>
5
6  Adafruit_PWMServoDriver pwm = Adafruit_PWMServoDriver();
7
8  // تعريف القنوات
9  #define BASE_CH 0
10 #define RIGHT_CH 1
11 #define LEFT_CH 2
12 #define CLAW_CH 3
13
14
15 int baseMin = 0, baseMax = 90;
16 int leftMin = 0, leftMax = 90;
17 int rightMin = 40, rightMax = 120;
18 int clawOpen = 60, clawClose = 90;
19
20 void setup() {
21   Serial.begin(9600);
22   pwm.begin();
23   pwm.setPWMFreq(60);
24
25   moveServo(BASE_CH, 0);
26   moveServo(LEFT_CH, 40);
27   moveServo(RIGHT_CH, 80);
28   moveServo(CLAW_CH, clawOpen);
29   delay(3000);
30 }
31
32 void loop() {
33   // --- 1. التوجه نحو الغرض ---
34   moveServo(BASE_CH, 45);
35   delay(2000);
36
37   // --- 2. النزول لالتقاط الغرض ---
38   moveServo(LEFT_CH, 70); // نزول ضمن حد (90-0)
39   moveServo(RIGHT_CH, 80); // ميلان ضمن حد (120-40)
40   delay(2000);
41
42   // --- 3. إغلاق المخلب ---
43   moveServo(CLAW_CH, clawClose);
44   delay(2000);
45
46   // --- 4. الرفع للأعلى ---
47   moveServo(LEFT_CH, 20);
48   moveServo(RIGHT_CH, 110);
49   delay(2000);
50
51   // --- 5. الدوران لنقل الغرض ---
52   moveServo(BASE_CH, 90);
53   delay(2500);
54
55   // --- 6. النزول لوضع الغرض ---
56   moveServo(LEFT_CH, 60);
57   moveServo(RIGHT_CH, 90);
58   delay(2000);
59
60   // --- 7. فتح المخلب (الإفلات) ---
61   moveServo(CLAW_CH, clawOpen);
62   delay(1500);
63
64   // --- 8. الرفع للأعلى ---
65   moveServo(LEFT_CH, 20);
66   moveServo(RIGHT_CH, 110);
67   delay(2000);
68
69   // --- 9. العودة لوضعية الاستعداد ---
70   moveServo(LEFT_CH, 40);
71   moveServo(RIGHT_CH, 80);
72   moveServo(BASE_CH, 0);
73
74   Serial.println("تم الانتهاء من الدورة.. انتظر 5 ثوان");
75   delay(5000);
76 }
77
78 // دالة التحريك مع الحماية (Constrain)
79 void moveServo(uint8_t channel, int angle) {
80   int targetAngle = angle;
81
82   if (channel == BASE_CH) targetAngle = constrain(angle, baseMin, baseMax);
83   if (channel == LEFT_CH) targetAngle = constrain(angle, leftMin, leftMax);
84   if (channel == RIGHT_CH) targetAngle = constrain(angle, rightMin, rightMax);
85   if (channel == CLAW_CH) targetAngle = constrain(angle, clawOpen, clawClose);
86
87   int pulse = map(targetAngle, 0, 180, 150, 600);
88   pwm.setPWM(channel, 0, pulse);
89 }
```

Output

Output

• Results

The robotic arm successfully completed automated Pick-and-Place operations during testing. The system demonstrated:

- Reliable object Pickup and placement.
- Stable operation without controller rests.
- Consistent Performance during repeated cycles.
- Smooth servo movement through PWM control.



• Future Improvements

- Implementation of inverse kinematics for coordinate-based control.
- Integration of ultrasonic sensors for object detection.
- Bluetooth communication using an HC-05 module.
- Camera-based vision systems for autonomous object sorting.

- Conclusion

The Project successfully demonstrated 4-DoF robotic arm capable of automated Pick-and-place operations. Using Arduino control, PWM servo driving, and an external power supply, the system achieved reliable and stable performance.

- References

- 1- Arduino Uno R3 Documentation.
- 2- Adafruit PCA9685 Servo Driver Documentation.
- 3- Robot Modeling and control - M. W. Spong.
- 4- Arduino IDE Official Documentation.